LESSON 8.2

# QUOTIENT LAWS OF EXPONENTS

## LESSON INTRODUCTION

**MA.7.NSO.1.1** Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions limited to whole-number exponents and rational-number bases.

## LEARNING TARGETS

- I can explain and use the quotient of powers, power of a quotient, and zero exponent laws.

## BENCHMARK COHERENCE

**BUILDING ON**

**MA.6.NSO.3.3** Evaluate positive rational numbers with natural-number exponents.

**WORKING TOWARD**

**MA.8.AR.1.1** Apply the laws of exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

**MA.8.NSO.1.3** Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expression, limited to integer exponents and rational-number bases, with procedural fluency.

## ESSENTIAL QUESTIONS

- How can the laws of exponents be used to write an equivalent expression?
- Why must the bases be the same to apply the quotient laws of exponents?

## LESSON NARRATIVE

In this lesson, students explore the quotient laws of exponents: quotient of powers, power of a quotient, and zero exponent laws. Students will engage in station-based collaboration for each of the laws of exponents to bring it all together at the end of the lesson to review formal definitions. After the collaboration activities, students will apply these laws to expressions that have rational bases.

## COMPONENT SUMMARY

**8.2.1 WARM-UP** *(~5 MIN)*

Describing patterns activity

**8.2.2 ACTIVITY** *(5-10 MIN)*

Quotient of powers station

**8.2.3 ACTIVITY** *(5-10 MIN)*

Power of a quotient station

**8.2.4 ACTIVITY** *(5-10 MIN)*

Zero exponent station

**8.2.5 BRING IT TOGETHER** *(5-10 MIN)*

Quotient laws of exponents with rational bases

**8.2.6 YOUR TURN!** *(~10 MIN)*

Quotient laws of exponents practice

**8.2.7 WRAP-UP** *(~5 MIN)*

Self-reflection activity

# RESOURCES

### GLOSSARY

Key Terms:

- Quotient of powers exponent law
- Power of a quotient exponent law
- Zero exponent law

Additional Terms: N/A

### MATERIALS

- (Optional) Printed stations activities
- Calculators

### ADDITIONAL PREPARATION

- Consider how each station should look. The class can be broken into small groups or partner pairs, and there can be two sets of each station. Students can either rotate through the stations or the stations can be passed from group to group.

# TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may try to divide the exponents instead of subtracting.
- Students may only apply the exponent to the numerator of a fraction and not the denominator as well.
- Students may think that a base raised to the power of zero is zero.

### THINGS TO CONSIDER

- While exploring and discovering the different laws, student pairs may not come up with a rule that matches the law they should be discovering. During the activities, consider circulating to help guide partners to think of a helpful rule or asking questions to help them refine their rule.
- If time is an issue, then consider assigning 8.2.5 Bring It Together and 8.2.6 Your Turn! for homework.

### LITERACY AND LANGUAGE ROUTINES

**Clarify, Critique, Correct**

In each collaborative station activity, students will be given the opportunity to explore scenarios that will result in their discovery of quotient laws of exponents: quotient of powers, power of a quotient, and zero exponent laws. When closing each activity, consider leveraging this protocol to have students critique their findings and correct based on feedback and the formal definition, when introduced.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

**MA.K12.MTR.4.1 Discussion and Reflection**

This lesson contains three collaborative stations where students will work collaboratively with a partner to discover the quotient laws of exponents: quotient of powers, power of a quotient, and zero exponent laws. Through these activities, students will engage in collaborative discussion and reflect on their findings when introduced to the formal definition of each law.

# DIFFERENTIATE INSTRUCTION

Consider keeping the same partner pairings that were established in Lesson 1. The content is different, but the structure is similar, making the entry point for existing partnerships more consistent.

Consider creating a foldable, Frayer Model, or note-taking guide to keep a central location for these specific laws of exponents. This will allow students to refer back to them as they continue to practice throughout the unit.

Leverage students' performances on the collaborative stations to determine which students may need differentiated support during 8.2.6 Your Turn!. If students are struggling with the general concept, they will have a difficult time with being accurate in the practice activity.

# 8.2.1 WARM-UP: USE A PATTERN

**QUESTIONING**

For students who finish early, challenge them to write an expression for the number of squares in Figure n. ($n+n+2+n^2$)

**8.2.1 Warm-Up: Use a Pattern**

Look at the pattern shown in the figures below. Work with your partner to answer the questions.

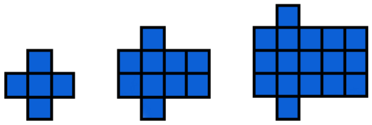


Figure 1 Figure 2 Figure 3

1. Describe the pattern in words.
   *Answers will vary. The first has 5 blocks, the second has 10 blocks, and the third has 17 blocks. Add 1 block to the first and second column and the remaining columns are the figure's number squared.*

2. If the pattern continued, how many squares would be in Figure 10?
   *Column 1 has 10 blocks, column 2 has 12 blocks, and $10^2 = 100$ blocks. The sum is 122 blocks.*

### 8.2.2 Activity: Station 1: Quotient of Powers Exponent Law

Work with your partner or a small group to complete three stations involving the laws of exponents.

**Task 1:**

1. Review the steps Rosa, Victor, and Emma each used to generate an equivalent expression. Then, answer the questions that follow.

| | Rosa | Victor | Emma |
|---|---|---|---|
| **Given** | $\frac{5^6}{5^3}$ | $\frac{10^4}{10^2}$ | $\frac{7^4}{7^1}$ |
| **Student Work** | $\frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5 \cdot 5}$<br>$5 \cdot 5 \cdot 5$ | $\frac{10 \cdot 10 \cdot 10 \cdot 10}{10 \cdot 10}$<br>$\boxed{\frac{10}{10}} \boxed{\frac{10}{10}} \frac{10}{1} \cdot \frac{10}{1}$<br>$1 \cdot 1 \cdot 10 \cdot 10$<br>$10 \cdot 10$ | $\frac{7 \times 7 \times 7 \times 7}{7}$<br>$7 \times 7 \times \boxed{\cancel{\frac{7}{7}}} \times 7$<br>$7 \times 7 \times 7$ |
| **Equivalent Expression** | $5^3$ | $10^2$ | $7^3$ |

a. What do you notice about the bases of the numerator and denominator of each given exponential expression?
*The bases are the same in the numerator and the denominator.*

b. Circle the exponents in the given exponential expression and circle the exponent in the equivalent expression each student created. How are they related?
*The exponents in the equivalent expressions are the difference between the exponents in the numerators and the denominators of the original expression.*

# 8.2.2 ACTIVITY: STATION 1: QUOTIENT OF POWERS EXPONENT LAW

## TEACHER MOVES

Consider how to utilize each activity for your class. The class can be broken into small groups or partner pairs, and there can be two sets of each station. Students can either rotate through the stations, or the stations can be passed from group to group. Circulate to ensure that students understand the law that was discovered before moving to the next exploration.

## ENRICHMENT

- Can the quotient be found without using the quotient of powers exponent law?
- What is the benefit of using the quotient of powers exponent law instead of writing the repeated multiplication steps and dividing?
- Why does the quotient of powers exponent law require the exponents to be subtracted and not divided?

## TEACHER MOVES

While circulating the room, listen for student responses to question 1a. As they begin to articulate what they notice, it would be a strong opportunity to hear if students are noticing the information about the bases and not a general noticing that won't lead to their formulation of a rule. It can also provide a glimpse into potential misconceptions that may need to be addressed.

# 8.2.2 ACTIVITY: STATION 1: QUOTIENT OF POWERS EXPONENT LAW (CONTINUED)

## DIFFERENTIATE INSTRUCTION

Allow students to add these new exponent laws to their existing foldable, guided-notes reference sheets, additional Frayer Models, etc. Students should have a single exponent law reference document that they can use throughout the unit.

## TEACHER MOVES

For task 3, pay close attention to what students are doing with the negative sign. Oftentimes, a common misconception is for students to not evaluate the base with the negative sign when there is an exponent. You can tell if this is happening if a student's answers to questions 1 and 3 are positive versions of the correct answer.

**Task 2:**

1. Fill in the blank to complete the rule for dividing exponential expressions with the same bases.

When dividing two powers that have the same nonzero base, keep the base and ___*subtract*___ the exponents. This is called the **quotient of powers exponent law**.

2. When $a$ is a nonzero real number, and $m$ and $n$ are integers, the exponential expression $\frac{a^m}{a^n}$ is equal to

- ○ $a^{mn}$.
- ○ $a^{m-n}$.
- ○ $a^{m+n}$.

**Task 3:** Show your steps of applying the quotient of powers exponent law to evaluate.

1. $\frac{(-2)^6}{(-2)^3} = (-2)^{6-3} = (-2)^3 = -8$

2. $\frac{(3)^6}{(3)^2} = 3^{6-2} = 3^4 = 81$

3. $\frac{(-7)^4}{(-7)^1} = (-7)^{4-1} = (-7)^3 = -343$

**8.2.3 Activity: Station 2: Power of a Quotient Exponent Law**

**Task 1:** Luis and Maya have shown their steps for using the power of a quotient exponent law to create equivalent expressions. Amir is confused. Help Amir complete his task by showing similar steps to Luis and Maya for his expression $\left(\frac{7}{3}\right)^4$.

| Luis | Maya | Amir |
|---|---|---|
| $\left(\frac{8}{10}\right)^3$ | $\left(\frac{3}{11}\right)^5$ | $\left(\frac{7}{3}\right)^4$ |
| $\frac{8}{10}\cdot\frac{8}{10}\cdot\frac{8}{10}$ | $\frac{3}{11}\cdot\frac{3}{11}\cdot\frac{3}{11}\cdot\frac{3}{11}\cdot\frac{3}{11}$ | $\frac{7}{3}\cdot\frac{7}{3}\cdot\frac{7}{3}\cdot\frac{7}{3}$ |
| $\frac{8\cdot 8\cdot 8}{10\cdot 10\cdot 10}$ | $\frac{3\cdot 3\cdot 3\cdot 3\cdot 3}{11\cdot 11\cdot 11\cdot 11\cdot 11}$ | $\frac{7\cdot 7\cdot 7\cdot 7}{3\cdot 3\cdot 3\cdot 3}$ |
| $\frac{8^3}{10^3}$ | $\frac{3^5}{11^5}$ | $\frac{7^4}{3^4}$ |

**Task 2:**

1. Fill in the blank to complete the rule for raising a fraction to a power.

When raising a fraction to a power, the result is equal to the quotient of the numerator and denominator raised separately to the *same* power. This is called the **power of a quotient exponent law**.

2. When $a$ and $b$ are nonzero real numbers, and $m$ is an integer, the exponential expression $\left(\frac{a}{b}\right)^m$ is equal to

- ○ $a^{b-m}$.
- ○ $\frac{a^m}{b^m}$.
- ○ $a^m - b^m$.

**Task 3:** Apply the power of quotient exponent law. Do not simplify the fractions.

1. $\left(\frac{-4}{8}\right)^2 = \frac{(-4)^2}{8^2}$

2. $\left(\frac{-4}{-2}\right)^6 = \frac{(-4)^6}{(-2)^6}$

3. $\left(\frac{2}{-8}\right)^5 = \frac{2^5}{(-8)^5}$

# 8.2.3 ACTIVITY: STATION 2: POWER OF A QUOTIENT EXPONENT LAW

## TEACHER MOVES

The names of the laws are very similar (using "product" and "power" in different orders). Students may grasp the concepts for each one but articulate the name of the law incorrectly or in reverse order. Try not to penalize the students, but rather, remind them that using the correct vocabulary is important when explaining conclusions and engaging in mathematical discourse. A class anchor chart may be helpful for students as they are learning the names.

## DIFFERENTIATE INSTRUCTION

Allow students to add the power of a quotient exponent law to their existing foldable or alternative reference document. Consider having students highlight the words "power" and "quotient" in different colors to see the order of the phrasing and how that connects to their notes.

## TEACHER MOVES

Consider reviewing the instructions for task 3. While it states "do not simplify the fractions," many student pairs will try to do so anyway. Place an additional example on the board such as $(-\frac{5}{8})^3$ and show them how the task is simply asking them to apply the power of quotient exponent law without simplifying, to produce an answer such as $\frac{(-5)^3}{8^3}$.

# 8.2.4 ACTIVITY: STATION 3: ZERO EXPONENT LAW

## TEACHER MOVES

A common misconception is for students to think that Hannah is right and Sebastian is wrong because their answers appear to be different. This misconception prevents students from making the connection to the zero exponent law. Use discussion to help solidify this concept for students without giving them the formal definition yet.

### 8.2.4 Activity: Station 3: Zero Exponent Law

**Task 1:** Two students used their understanding of exponents to write an equivalent expression for the exponential expression $\frac{2^3}{2^3}$.

1. Discuss each step that the students completed.

| Hannah | Sebastian |
|---|---|
| $\frac{2^3}{2^3}$ | $\frac{2^3}{2^3}$ |
| $\frac{2 \times 2 \times 2}{2 \times 2 \times 2}$ | $2^{3-3}$ |
| $\frac{2}{2} \times \frac{2}{2} \times \frac{2}{2}$ | $2^0$ |
| $1 \times 1 \times 1$ | |
| $1$ | |

2. Using Hannah's and Sebastian's work, make a prediction about the value of any number raised to the power of 0.
*Answers will vary. All non-zero numbers raised the zero power equal one; any non-zero number raised to the zero power is one.*

## ENRICHMENT

- Why do you think the pattern exists the way that it does for each row in the table of Task 2?
- Do you think your answer for question 4 could be used for any number?
- Using the completed table, why do you think that a number raised to the power of 0 would be 1 and not 0?
- For question 4: Would this pattern hold true if the base were 0? What about the pattern for 0 makes this different?

## DIFFERENTIATE INSTRUCTION

Allow students to add the zero exponent law to their existing foldable or alternative reference document.

**Task 2:** Use the tables to answer the questions.

| **Exponent Form** | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|---|
| **Value** | 16 | 8 | 4 | 2 | 1 |

| **Exponent Form** | $3^4$ | $3^3$ | $3^2$ | $3^1$ | $3^0$ |
|---|---|---|---|---|---|
| **Value** | 81 | 27 | 9 | 3 | 1 |

1. What pattern do you see in the top row of each table?
*Answers will vary. The base stays the same and the exponent is being decreased by one each time.*

2. What pattern do you see in the bottom row of each table?
*Answers will vary. In the first table the numbers in the bottom row are divided by two and in the second table, it is being divided by three.*

3. How is the pattern in the bottom row of each table related to the value of the base in the top row of the table?
*Answers will vary. Each time the number in the bottom row is divided by the base, the value of the exponent in the row above decreases by one.*

4. Use the patterns in the tables to find the values of $2^0$ and $3^0$. Fill in the empty cell in each table.

$2^0 =$ __1__

$3^0 =$ __1__

**Task 3:** Fill in the blank to complete the rule for a base with a zero exponent.

When a nonzero real number is raised to the zero power, $a^0$, its value equals _*one*_. This is called the **zero exponent law**.

### 8.2.5 Bring It Together: Exponential Expressions with Rational Bases

1. Match each set of expressions with the Exponent Law that it demonstrates. Use arrows to indicate matches.

| | |
|---|---|
| $\left(\frac{0.5}{1.9}\right)^3$<br>$\left(\frac{0.5}{1.9}\right) \cdot \left(\frac{0.5}{1.9}\right) \cdot \left(\frac{0.5}{1.9}\right)$<br>$\frac{0.5 \cdot 0.5 \cdot 0.5}{1.9 \cdot 1.9 \cdot 1.9}$<br>$\frac{0.5^3}{1.9^3}$ | Zero Exponent Law |
| $\left(\frac{2}{3}\right)^0$<br>$1$ | Quotient of Powers Law |
| $\frac{10.1^6}{10.1^4}$<br>$\frac{10.1 \cdot 10.1 \cdot 10.1 \cdot 10.1 \cdot 10.1 \cdot 10.1}{10.1 \cdot 10.1 \cdot 10.1 \cdot 10.1}$<br>$\frac{10.1}{10.1} \cdot \frac{10.1}{10.1} \cdot \frac{10.1}{10.1} \cdot \frac{10.1}{10.1} \cdot \frac{10.1}{1} \cdot \frac{10.1}{1}$<br>$1 \cdot 1 \cdot 1 \cdot 1 \cdot 10.1 \cdot 10.1$<br>$10.1^2$ | Power of a Quotient Law |

2. Complete the following steps for each expression.

- Rewrite each expression shown by using the laws of exponents.
- Check off the exponent law used from the list.
- Evaluate the exponential expression.

| | |
|---|---|
| $\frac{0.5^8}{0.5^5}$ | ○ Quotient of powers law<br>○ Power of quotient law<br>○ Zero exponent law |
| $\frac{0.5^8}{0.5^5} = 0.5^3 = 0.125$ | |

| | |
|---|---|
| $\left(\frac{0.2}{10}\right)^4$ | ○ Quotient of powers law<br>○ Power of quotient law<br>○ Zero exponent law |
| $\left(\frac{0.2}{10}\right)^4 = \frac{0.2^4}{10^4} = 0.00000016$ | |

| | |
|---|---|
| $\left(\frac{3}{4}\right)^0$ | ○ Quotient of powers law<br>○ Power of quotient law<br>○ Zero exponent law |
| $\left(\frac{3}{4}\right)^0 = 1$ | |

## 8.2.5 BRING IT TOGETHER: EXPONENTIAL EXPRESSIONS WITH RATIONAL BASES

### TEACHER MOVES

While this activity is primarily about identifying which exponent law is being applied, it is important that attention is given to the concept of rational bases. Often, students think that because the base is not a whole number, it becomes a different problem altogether. Position students to see that even with rational bases, the process for applying the exponent law is the same but the calculations will be different. For question 1, students are only expected to match the exponent law to the correct expression set. Be mindful that the first set requires the use of two laws. Help students see that it is okay for that to occur, since they may think that it is always limited to one.

### TEACHER MOVES

The values of the first two expressions can be determined by hand. However, there are a lot of activities in this lesson, so for the purposes of time, consider allowing students to use a calculator when determining the value of the final expression.

### QUESTIONING

For a challenge, consider adding in $0^0$ to see what students think the value would be. Allow them to use a calculator to verify what they think would happen. If time allows, this can create a powerful discussion.

## 8.2.6 YOUR TURN!

### TEACHER MOVES

Remind students that the equivalent column of the table is designed to be after you've applied the exponent law and that fractions do not have to be simplified. Refer them back to 8.2.3 Activity, Task 3. Not providing access to calculators will help students focus on applying the law, not evaluating.

### DIFFERENTIATE INSTRUCTION

The second column of the table has students identify the exponent law. If students get this wrong but the expression right, it can indicate a memory or confusion issue with naming.

If students get the exponent law right but the expression wrong, it could indicate a calculation error or switching the process with another law in error.

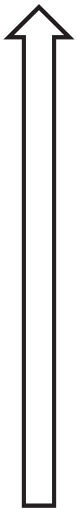
**8.2.6 Your Turn!**

1. Rewrite the expressions using laws of exponents. State the exponent law used.
   *Answers could vary.*

| Expression | Equivalent | Exponent Law |
|---|---|---|
| $\frac{8.6^6}{0.2^6}$ | $\left(\frac{8.6}{0.2}\right)^6$ | *Power of quotient* |
| $\frac{15.7}{15.7}$ | $\frac{15.7^1}{15.7^1} = 15.7^0 = 1$ | *Zero exponent* |
| $\frac{6.75^7}{6.75^3}$ | $6.75^4$ | *Quotient of powers* |
| $\left(\frac{5}{20}\right)^{10}$ | $\frac{5^{10}}{20^{10}}$ | *Power of quotient* |
| $\left(\frac{0.85}{2.64}\right)^7$ | $\frac{0.85^7}{2.64^7}$ | *Power of quotient* |
| $8.45^0$ | $1$ | *Zero exponent* |

**8.2.7 Wrap-Up: Reflection**

1. Shade the arrow up to the statement that best describes your current level of understanding.
*Answers will vary.*



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will make sure that I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

# 8.2.7 WRAP-UP: REFLECTION

**DIFFERENTIATE INSTRUCTION**

As an alternative, consider having students go through the lesson and place a number on each section to indicate their level of understanding.

- 1 being "I do not understand any of this yet."
- 2 being "I understand some of this."
- 3 being "I completely understand this."